SOKOINE UNIVERSITY OF AGRICULTURE

SCHOOL OF ENGINEERING AND TECHNOLOGY (SoET)

DEPARTMENT OF AGRICULTURAL ENGINEERING (DAE)

AE 219

ASSIGNMENT 3

Note:

[1] Please listen to videos https://www.youtube.com/playlist?list=PLmKnXyyknMkPhutqZ6uebPazrUCobJ8Ep provided by my Lab YEESI Lab and instructed by Dr Masamaki

[2] Please use Google Colab: https://colab.research.google.com/

[3] Please read and listen to the videos in https://towardsdatascience.com/best-time-saving-python-tricks-in-jupyter-notebook-fca877132507

Questions:

1. Write a Python program to calculate and display the sum of the series (1*1) + (1*2) + (1*2*3) + (1*2*3*4) + (1*2*3*4*5) + ... + (1*2*3*4*5….*n). Assume n=40. Validate **your program using Google Colab and submit it as a Jupyter Notebook program in the iPynb extension with comments.**
2. Write a Python program to calculate and display the sum of the series 1*2/1! + 2*3/2! + 3*4/3! + 4*5/4! +…….+ n*(n+1)/n!. Assume ! means factorial and n=40. **Validate your program using Google Colab and submit it as a Jupyter Notebook program in the iPynb extension with comments.**
3. Navigate and download the test dataset found in the following link: https://zindi.africa/competitions/tanzania-tourism-prediction/data
   Then, write a Python program to calculate the correlation between the column total_males and total_females. **Validate your program using Google Colab and submit it as a Jupyter Notebook program in the iPynb extension with comments.**

   Submit your code results by Saturday midnight (17-June-2023) to the CR as the softcopy and CR will forward the bundle of all codes to me by Sunday. DO NOT SEND YOUR WORK to the Instructor's email. For any enquiries, please send an email.

   As usual, Every one hour late equates to 5 marks deduction.

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ASSIGNMENT 5

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[4] Please read how to input data into a program: https://www.askpython.com/python/examples/python-user-input

Questions (All programs should run on the command line. Read Instruction 4):

1. **Write a Python program to find the transpose of a matrix (interchange row & column)**
   1.1.Allow the user to Enter the order of the matrix
   1.2.Allow user to Enter the co-efficient of the matrix
   1.3.Display both matrix and its transpose
2. **Write Python program to convert binary to decimal.**
   2.1.Allow user to Enter the binary number(1s and 0s)
   2.2.Display binary and its' decimal equivalent
3. **Write Python program to check the equality of two matrix**
   3.1.Allow user to enter the order of the matrix A and B
   3.2.Check the two if they are equal
4. **Write Python Program to Concatenate 2 strings without strcat function**
   4.1.Allow user to Enter the first and second string then concatenate them without using inbuilt strcat
   4.2.Display concatenated string

5. **Write a Python program to find vowels and consonants in the given sentence**
   - ***5.1.*** *Allow user to Enter a sentence*
   - ***5.2.*** *Calculate and Display No. of vowels and consonants and then display the No. of vowels and consonants.*
6. **Write a Python program to generate and print the first N numbers of the following series 1, 2, 5, 9, 16, 27, 45…**
   - ***6.1.*** *Allow the user to Enter N so as to specify how long the series should be. E.g. If the user specifies 4 then the program should print only 1, 2, 5, 9. Note that the user should only be allowed to request a series of more than 3 elements.*
7. **Write a Python program to process and display hourly-paid cleaning employees' payroll for weekly payments. Use Python's in-built data structure like a list to hold your data variables. Assume the program processes payments for one week only**
   - ***7.1.*** The program should accept names, addresses and ages of only 30 employees {Name, address, and age}. The program should only accept employees who are 21 years old or more. [
   - ***7.2.*** The program should accept hours worked for each employee every day for seven days from Monday to Sunday. Employees are allowed to work for more than 8 hours per day. Extra time is paid separately.
   - ***7.3.*** The program should calculate the weekly salary for each employee. Assume every hour the worker receives Tsh 4,000 while for an overtime hour is Tsh 6000.
   - ***7.4.*** The program should display the names of cleaning employees and weekly salaries in descending order of the salary.
   - ***7.5.*** The program should be able to show/display the best day in which most employees accumulated the most hours
   - ***7.6.*** Note that >> The program should have a menu to allow the employer to do the above five tasks

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[4] Please read how to input data into a program: https://www.askpython.com/python/examples/python-user-input

Questions (All programs should run on the command line. Read Instruction 4):

1. **Write a Python program to process and display Wireless Sensor Network (WSN) data installed at SUA Model Farm weekly. Use Python in-built data structures like a dictionary, list etc., to hold your data variables. Assume the program processes sensor data for one week only**

   1.1. The program should accept names and locations (longitude and Latitude) for only 30 sensors {Name, location }. *The program should only take sensors located in the SUA model farm.*

   1.2. The program should accept data (R_n net radiation at the crop surface [MJ m^{-2} day^{-1}], G soil heat flux density [MJ m^{-2} day^{-1}], T air temperature at 2 m height [°C], u_2 wind speed at 2 m height [m s^{-1}], e_s saturation vapour pressure [kPa], e_a actual vapour pressure [kPa], γ psychrometric constant [kPa $°C^{-1}$].) every hour every day for seven days, from Monday to Sunday. Sensor data should be collected in daylight only from 7:00 AM to 6:00 PM.

   1.3. The program should calculate each sensor's daily ETo reference evapotranspiration [mm day-1]. See the formula for Δ slope vapour pressure curve [kPa $°C^{-1}$] below.

$$ET_o = \frac{0.408\Delta(R_n - G) + \gamma \frac{900}{T+273} u_2 (e_s - e_a)}{\Delta + \gamma(1 + 0.34u_2)}$$

$$\Delta = \frac{4098\left[0.6108\exp\left(\frac{17.27T}{T+237.3}\right)\right]}{(T+237.3)^2}$$

1.4. The program should display the names of sensors and daily reference evapotranspiration in descending order of the daily reference evapotranspiration.

1.5. The program should be able to show/display the day in which most of the sensors had the largest reference evapotranspiration.

1.6. Note that >> The program should have a menu to allow the Model farm Technician to do the above five tasks

2. **Write a Python program to process and display the Normalized Difference Vegetation Index (NDVI) plant values of the Rice Farm located in SUA Model Farm. A technician chose 20 plants randomly on the rice farm and collected the Red and Near-infrared (NIR) reflectance signature using a Multispectral sensor on the plants' leaves to measure crop vigour for precision fertilization. Use Python in-built data structures like lists or dict to hold your data variables. Assume the technician collected data every week for 16 weeks only.**

2.1. The program should accept plant heights, widths and locations (longitude and Latitude) for only 20 plants {height, width, location }. *The program should only take plants located on the SUA model rice farm.*

2.2. The program should accept plant data (red and near-infrared reflectance) every week for 16 weeks.

2.3. The program should calculate the weekly NDVI for each plant. Use the NDVI formula below:

$$NDVI = \frac{(NIR - Red)}{(NIR + Red)}$$

2.4. The program should display the locations, heights, widths and NDVI of the chosen plants in descending order of the NDVI.

2.5. The program should be able to show/display the best day when most plants had high NDVI.

2.6. Note that >> The program should have a menu to allow the technician to do the above five tasks

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